ASOCIACIÓN ASTRONÓMICA
Y EDUCATIVA DE CANARIAS
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# E5: Introduction to orbits and black holes

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## Worksheet 3: Introduction to black holes

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# 1. Introduction

In this final worksheet, you will perform a series of fundamental analytical calculations regarding black holes using the Schwarzschild metric as a starting point. This solution describes the geometry of space-time outside a spherically symmetric body of mass $M$ at rest. The function characterizing the Schwarzschild solution is given by:

$$f(r) = 1 - \frac{2GM}{rc^2},$$

where $G = 6.674 \times 10^{-11}\ N \cdot \frac{m^2}{kg^2}$ and $c = 3 \times 10^8\ m/s$. In this worksheet, we will momentarily step away from the Python environment to focus your attention on direct mathematical analysis.

# 2. The Density of a Black Hole

The Schwarzschild radius ($r_s$) defines the event horizon: that invisible boundary beyond which nothing, not even light, can escape. It is given by

$$r_s = \frac{2GM}{c^2}.$$

1. What is the Schwarzschild radius of the Sun and the Earth ($R_\odot = 7 \times 10^5\ km,\ M_\odot = 2 \times 10^{30}\ kg,\ M_\oplus = 6 \times 10^{24} kg, R_\oplus = 6 \times 10^3 km$)? How does the Schwarzschild radius compare to the actual radii of the Sun and the Earth? For what values of the radius is the Schwarzschild solution valid in these cases?
2. Neutron stars are the densest stars in the universe and one of the precursors to black holes. If a neutron star exceeds a mass of about $3M_\odot$, its final fate is to collapse into a black hole. Assuming a radius of $10\ km$ and a mass of $1.4M_\odot$, calculate its Schwarzschild radius and its density.
3. Imagine we have a black hole whose Schwarzschild radius is $r_s = 10\ km$. What is its mass?

After having performed all these calculations, what would happen to the orbits of the solar system if we suddenly replaced the Sun with a black hole of the same mass?

# 3. Tidal Forces

The fascinating effects of black holes occur when we are very close to their event horizon. Among them is the effect known as spaghettification: if we were swallowed by a black hole of a few solar masses, as we approached its horizon, we would be stretched from head to toe, turning into a *spaghetti.*

This effect occurs due to tidal forces. Gravity varies with respect to the distance from the attracting body in both Newtonian and Einsteinian physics. When considering a very large body, or an extreme change in gravity over a very small space, there can be considerable differences in acceleration between two points of the same body. This effect is known as *tidal acceleration,* or if we are thinking of forces instead, *tidal forces.* The difference in the Moon's gravitational pull on different points of our planet is the main contributor to tides, hence the name. If the tidal forces acting on a body are greater than the forces holding it together, that body will deform and, in extreme cases, be torn apart.

Both Newton and Einstein agree that the difference in acceleration experienced by two points at a distance $d$, located at a distance $r$ (with $r \gg d$) from an attracting body of mass $M$ is given by

$$\Delta a = \frac{2\,G\,M\,d}{r^3}.$$

1. Calculate the difference in acceleration between your head and your feet due to being on the Earth's surface.
2. What is the difference in acceleration caused by the Sun's attraction on the Earth?
3. Suppose a black hole has the mass of the Sun and a Schwarzschild radius of about 2.5 $km$. What is the tidal acceleration we would experience at about 100 km? Would we be spaghettified?
4. A supermassive black hole has a mass of about 100 million times the mass of the Sun and a Schwarzschild radius of about 295 million kilometers. What would be the tidal acceleration experienced 100 km from the event horizon?

# 4. Orbits of Light

Until now, we have concentrated on orbits of bodies with mass. Another major triumph of General Relativity is that it also describes how massless light interacts with gravity. This is a sneak peek at beautiful, more formal results.

General Relativity is not about forces, but about trajectories known as geodesics. The movement of light in space, specifically within the Schwarzschild metric, is determined by a type of geodesic known as a *null geodesic*. These trajectories are characteristic of light and can be curved due to gravity. For the Schwarzschild metric, null geodesics can be described by the effective potential

$$V_{eff}(r) = \frac{ml^2}{2r^2}\left(1 - \frac{2k}{rc^2}\right).$$

Here $l$ is the angular momentum per unit mass, and $k = GM$, where $G = 6.674 \times 10^{-11}\ N \cdot \frac{m^2}{kg^2}$. Let us study this function.

1. What happens to the function as $r \to 0$? And for $r \to \infty$?
2. At what radius is there a local maximum? What do you think it represents?
3. Compare the last stable orbit of light with that of an object with mass.